

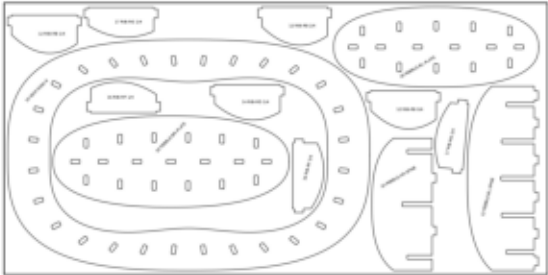
PEBBLE SOFA / CNC NOTES

4 sheets 2440 x 1220 x 18 mm (sheet size ASSUMED)

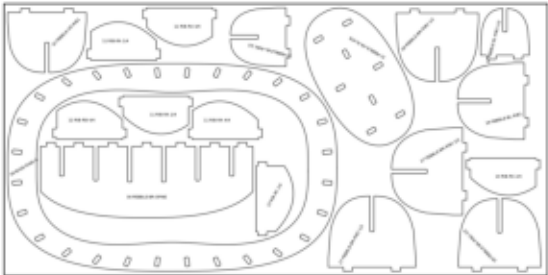
Units / scale	millimetres, 1:1 - never rescale the DXF
Layers	CUT (all closed, cut through) / ENGRAVE-LABEL (0.5-1 mm) / REFERENCE-SHEET (do not cut)
Joints	through-mortises 41 x 19 mm for 40 mm tabs in 18 mm board - UNTESTED FIT
Corner relief	R3.2 dogbone in every inside corner, sized for a 6 mm bit (ASSUMED)
Cutter offset	apply the normal tool-radius offset on every CUT contour; no extra compensation inside slots
Cut order	engrave, then inside contours (mortises, openings), then outlines
Hold-down	parts 10 mm apart, 9 mm clamp edge; add onion-skin or tabs for small ribs
Small parts	spacers and arches sit next to long rails - add tabs or onion-skin so they cannot move on the last pass

PEBBLE SOFA / CUTTING LAYOUT

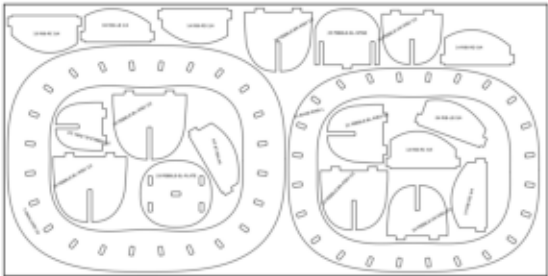
SHEET 1/4 - 2440 x 1220 x 18 mm MDF



SHEET 2/4 - 2440 x 1220 x 18 mm MDF



SHEET 3/4 - 2440 x 1220 x 18 mm MDF



SHEET 4/4 - 2440 x 1220 x 18 mm MDF



PEBBLE SOFA / VALIDATION

automated checks - geometry, joints, 3D assembly

PASS	[1] part topology	01 BASE-RING-L: 26 loops, simple, holes inside and apart
PASS	[1] part topology	02 SEAT-RING-L: 26 loops, simple, holes inside and apart
PASS	[1] part topology	03 RIB-LA: 1 loops, simple, holes inside and apart
PASS	[1] part topology	04 RIB-LB: 1 loops, simple, holes inside and apart
PASS	[1] part topology	05 RIB-LC: 1 loops, simple, holes inside and apart
PASS	[1] part topology	06 RIB-LD: 1 loops, simple, holes inside and apart
PASS	[1] part topology	07 RIB-LE: 1 loops, simple, holes inside and apart
PASS	[1] part topology	08 RIB-LF: 1 loops, simple, holes inside and apart
PASS	[1] part topology	09 BASE-RING-R: 30 loops, simple, holes inside and apart
PASS	[1] part topology	10 SEAT-RING-R: 30 loops, simple, holes inside and apart
PASS	[1] part topology	11 RIB-RA: 1 loops, simple, holes inside and apart
PASS	[1] part topology	12 RIB-RB: 1 loops, simple, holes inside and apart
PASS	[1] part topology	13 RIB-RC: 1 loops, simple, holes inside and apart
PASS	[1] part topology	14 RIB-RD: 1 loops, simple, holes inside and apart
PASS	[1] part topology	15 RIB-RE: 1 loops, simple, holes inside and apart
PASS	[1] part topology	16 RIB-RF: 1 loops, simple, holes inside and apart
PASS	[1] part topology	17 RIB-RG: 1 loops, simple, holes inside and apart
PASS	[1] part topology	18 PEBBLE-BL-PLATE: 17 loops, simple, holes inside and apart
PASS	[1] part topology	19 PEBBLE-BL-RIB1: 1 loops, simple, holes inside and apart
PASS	[1] part topology	20 PEBBLE-BL-RIB2: 1 loops, simple, holes inside and apart
PASS	[1] part topology	21 PEBBLE-BL-RIB3: 1 loops, simple, holes inside and apart
PASS	[1] part topology	22 PEBBLE-BL-SPINE: 1 loops, simple, holes inside and apart
PASS	[1] part topology	23 PEBBLE-EL-PLATE: 6 loops, simple, holes inside and apart
PASS	[1] part topology	24 PEBBLE-EL-RIB1: 1 loops, simple, holes inside and apart
PASS	[1] part topology	25 PEBBLE-EL-SPINE: 1 loops, simple, holes inside and apart
PASS	[1] part topology	26 PEBBLE-BR-PLATE: 20 loops, simple, holes inside and apart
PASS	[1] part topology	27 PEBBLE-BR-RIB1: 1 loops, simple, holes inside and apart
PASS	[1] part topology	28 PEBBLE-BR-RIB2: 1 loops, simple, holes inside and apart
PASS	[1] part topology	29 PEBBLE-BR-RIB3: 1 loops, simple, holes inside and apart
PASS	[1] part topology	30 PEBBLE-BR-SPINE: 1 loops, simple, holes inside and apart
PASS	[1] part topology	31 PEBBLE-AR-PLATE: 9 loops, simple, holes inside and apart
PASS	[1] part topology	32 PEBBLE-AR-RIB1: 1 loops, simple, holes inside and apart
PASS	[1] part topology	33 PEBBLE-AR-RIB2: 1 loops, simple, holes inside and apart
PASS	[1] part topology	34 PEBBLE-AR-SPINE: 1 loops, simple, holes inside and apart
PASS	[2] joints (Tokyo rule: 19 x 41 mortise for 18 mm board, 40 mm tab)	mortise 41 x 19 mm, tab 40 mm, board 18 mm
PASS	[2] joints (Tokyo rule: 19 x 41 mortise for 18 mm board, 40 mm tab)	152 mortises, all 19 x 41 mm: {(19.0, 41.0): 152}
PASS	[2] joints (Tokyo rule: 19 x 41 mortise for 18 mm board, 40 mm tab)	relief R3.2 clears a 6 mm cutter
PASS	[3] corner relief and cutter reach	01 BASE-RING-L: 0 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[3] corner relief and cutter reach	02 SEAT-RING-L: 0 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[3] corner relief and cutter reach	03 RIB-LA: 4 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[3] corner relief and cutter reach	04 RIB-LB: 4 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[3] corner relief and cutter reach	05 RIB-LC: 4 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[3] corner relief and cutter reach	06 RIB-LD: 4 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[3] corner relief and cutter reach	07 RIB-LE: 4 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[3] corner relief and cutter reach	08 RIB-LF: 4 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[3] corner relief and cutter reach	09 BASE-RING-R: 0 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[3] corner relief and cutter reach	10 SEAT-RING-R: 0 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[3] corner relief and cutter reach	11 RIB-RA: 4 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[3] corner relief and cutter reach	12 RIB-RB: 4 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[3] corner relief and cutter reach	13 RIB-RC: 4 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2

PEBBLE SOFA / VALIDATION (cont.)

PASS	[3] corner relief and cutter reach	14 RIB-RD: 4 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[3] corner relief and cutter reach	15 RIB-RE: 4 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[3] corner relief and cutter reach	16 RIB-RF: 4 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[3] corner relief and cutter reach	17 RIB-RG: 4 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[3] corner relief and cutter reach	18 PEBBLE-BL-PLATE: 0 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[3] corner relief and cutter reach	19 PEBBLE-BL-RIB1: 6 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[3] corner relief and cutter reach	20 PEBBLE-BL-RIB2: 6 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[3] corner relief and cutter reach	21 PEBBLE-BL-RIB3: 6 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[3] corner relief and cutter reach	22 PEBBLE-BL-SPINE: 22 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[3] corner relief and cutter reach	23 PEBBLE-EL-PLATE: 0 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[3] corner relief and cutter reach	24 PEBBLE-EL-RIB1: 6 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[3] corner relief and cutter reach	25 PEBBLE-EL-SPINE: 6 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[3] corner relief and cutter reach	26 PEBBLE-BR-PLATE: 0 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[3] corner relief and cutter reach	27 PEBBLE-BR-RIB1: 6 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[3] corner relief and cutter reach	28 PEBBLE-BR-RIB2: 6 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[3] corner relief and cutter reach	29 PEBBLE-BR-RIB3: 6 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[3] corner relief and cutter reach	30 PEBBLE-BR-SPINE: 26 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[3] corner relief and cutter reach	31 PEBBLE-AR-PLATE: 0 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[3] corner relief and cutter reach	32 PEBBLE-AR-RIB1: 6 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[3] corner relief and cutter reach	33 PEBBLE-AR-RIB2: 6 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[3] corner relief and cutter reach	34 PEBBLE-AR-SPINE: 10 corners relieved, worst spot a 6 mm bit cannot reach 0.00 mm2
PASS	[4] nesting sheet 1: 13 parts, closest 14.0 mm apart, all inside the 9 mm edge	
PASS	[4] nesting sheet 2: 18 parts, closest 14.0 mm apart, all inside the 9 mm edge	
PASS	[4] nesting sheet 3: 20 parts, closest 14.0 mm apart, all inside the 9 mm edge	
PASS	[4] nesting sheet 4: 29 parts, closest 13.3 mm apart, all inside the 9 mm edge	
PASS	[4] nesting every part placed its full quantity (80 pieces)	
PASS	[5] labels 80 pieces labelled	
PASS	[5] labels every label sits on material, 4 mm from any cut	
PASS	[6] built DXF units = mm	
PASS	[6] built DXF 236 closed CUT profiles (expected 236)	
PASS	[6] built DXF 80 ENGRAVE-LABEL texts	
PASS	[6] built DXF no duplicate CUT profiles (0 found)	
PASS	[6] built DXF longest part 1626 mm fits the sheet	
PASS	[3D-1] interference 145 neighbouring pairs, 0 interpenetrate	
PASS	[3D-2] every mortise holds a tab 152/152 mortises filled	
PASS	[3D-3] load path module L: 24/24 body ribs bear on both rings	
PASS	[3D-3] load path module R: 28/28 body ribs bear on both rings	
PASS	[3D-3] load path pebble BL: plate on the seat ring, 5 ribs + spine on the plate, egg-crate engaged	
PASS	[3D-3] load path pebble EL: plate on the seat ring, 2 ribs + spine on the plate, egg-crate engaged	
PASS	[3D-3] load path pebble BR: plate on the seat ring, 6 ribs + spine on the plate, egg-crate engaged	
PASS	[3D-3] load path pebble AR: plate on the seat ring, 3 ribs + spine on the plate, egg-crate engaged	
PASS	[3D-4] assembly sequence: every piece comes down vertically 80 pieces lowered in order BASE > BODY > SEAT > PLATE > PRIB > SPINE, 0 blocked	
PASS	[3D-5] overall size frame 3039 x 1062 x 668 mm on z = 0	